

FLAT CREEK WINTER CHARACTERIZATION STUDY

2017-2018
METHODS & DATA SUMMARY
DATA DELIVERABLES



March 8, 2018

Prepared for

Flat Creek Water Improvement District
PO Box 2037
Jackson, WY 83001

Prepared by

ALDERENVIRONMENTAL
water | wetlands | ecological consulting

Jackson, WY alderenvironmental.com

TABLE OF CONTENTS

BACKGROUND & INTRODUCTION	1
DATA COLLECTION PARAMETERS & METHODS	1
PARAMETERS & SITES	1
<i>Appendix A</i>	1
METHODS.....	2
<i>Water Temperature (Continuous and Instantaneous)</i>	2
<i>Air Temperature</i>	4
<i>Stream Stage</i>	4
<i>Water Level Loggers/Pressure Transducers (Continuous)</i>	4
<i>Stream Discharge</i>	5
<i>Time-lapse Photographs</i>	5
<i>Spot Photos</i>	5
<i>Ice Formation/ Flooding</i>	5
<i>Weather</i>	6
DATA SUMMARY	6
ARCGIS GEODATABASE & ONLINE VIEWER.....	6
<i>ArcGIS Geodatabase</i>	8
<i>ArcGIS Online Viewer</i>	8
ICE FORMATION: ICE DAMS.....	10
SPOT PHOTOS.....	11
TIME-LAPSE PHOTOS	12
LANDOWNER PHOTOS.....	12
WATER TEMPERATURE (LOGGERS).....	12
AIR TEMPERATURE & PRECIPITATION	12
WATER LEVEL (LOGGERS)	12
STREAM DISCHARGE (INSTANTANEOUS).....	13
THAW WELLS	13
USGS GAGE DATA.....	13
APPENDIX A – MONITORING PARAMETERS AND SITES	14
SHEET 1 - MONITORING PARAMETERS & DATA.....	15
SHEET 2 - MONITORING SITES	16
FIGURE 1 - MONITORING SITES.....	17
FIGURE 2 - MONITORING SITES NORTH	18
FIGURE 3 - MONITORING SITES MIDDLE	19
FIGURE 4 - MONITORING SITES SOUTH.....	20

BACKGROUND & INTRODUCTION

The Flat Creek Water Improvement District (FCWID) engaged Alder Environmental LLC to document the physical characteristics of Flat Creek within the Town of Jackson within the FCWID boundaries during the winter of 2017-2018. Quantitative and qualitative data collection between November 2017 and February 2018 focused on water and air temperature, ice conditions, flooding and thaw well operations. These data were collected and organized to provide a better understanding of ice formation, substantial icing areas and the effectiveness of existing thaw wells. Similar creek monitoring data was collected during the winters of 2015-2016 and 2016-2017.

Deliverables are provided in the following formats and locations:

- paper and electronic (pdf) report summarizing data collection methods and monitoring sites information,
- digital format of all data (Excel, GIS, jpeg, PDF, Hobo and Ruskin files) on DVD located in the back cover of this document and
- digital format of GIS data in an ArcGIS Online Cloud-Based Map Viewer.

Data were collected and organized with the assumption that the end user will be an individual or organization knowledgeable in interpreting the data collected. The end user should also be capable of using preloaded data in an ArcGIS Online viewer with some guidance from an individual knowledgeable with ArcGIS geodatabases and ArcGIS Online.

DATA COLLECTION PARAMETERS & METHODS

Data collection sites, parameters and methods were developed through a collaborative effort between Alder Environmental, FCWID, Town of Jackson, landowners and Ed Kempema, Ph.D. (Geology and Geophysics Dept., Univ. of WY). The field data collection period was from November 12, 2017 to February 13, 2018. Weather and stream data from outside sources were downloaded after the data collection period ended.

Parameters & Sites

Details regarding data parameters and monitoring sites are provided in Appendix A sheets and figures.

Appendix A

Sheet 1 – Monitoring Parameters and Data

Sheet 2 – Monitoring Sites & Descriptions

Figure 1 – Monitoring Sites Overview

Figure 2 – North Monitoring Sites

Figure 3 – Middle Monitoring Sites

Figure 4 – South Monitoring Sites

Methods

Flat Creek winter characterization data were generated from loggers, time-lapse cameras, observations, instantaneous measurements and from online database sources from external agencies. The equipment used for data collection is provided in Sheet 1 in Appendix A. The following provides supplemental information on data collection methods used by Alder Environmental.

Fifteen (15) monitoring events were conducted to record and map observations. These monitoring events took place twice a week during prolonged atmospheric high pressure cold periods. The goal was to record conditions on 75% of creek length within the FCWID boundaries. Observations or monitoring events were triggered according to the following general guidelines:

- 2 consecutive days of air temp <32°F or
- 1 consecutive days of air temp <25°F or
- air temp <10°F or
- observed ice formation in creek.

Monitoring events would start early in the morning and would consist of visiting all staff gage sites in addition to walking four designated routes: R-1a, R-1, R-2, R-3 (Figures 1-4). The start location was alternated for each monitoring between upstream and downstream. A Juniper Mesa 2 tablet with *Collector for ArcGIS* mapping software was used to collect and record data during monitoring events. Data locations were marked within the mapping software based off the GPS location and using the 2017 Teton County IR 1-ft aerial imagery as a reference.

Water Temperature (Continuous and Instantaneous)

RBRsolo T temperature loggers were installed at thirteen (13) sites in Flat Creek on November 7-8, 2017 as described and shown in Appendix A. Continuous water temperature in degrees Celsius was measured and logged every 5 minutes. Prior to deployment in the creek, the loggers were immersed in an ice bath to determine variability between loggers and accuracy of individual loggers. Methods for the ice bath are provided in the data spreadsheet (WaterTempLoggerData_Exported_IceBath_2017-2018.xlsx). The loggers were protected with a galvanized steel conduit attached by a chain to an anchor placed mid-channel on the bed of the creek. The anchor was connected to a fence t-post on the bank. Photos 1 and 2 below provide a visual of the deployment method.

Temperature loggers were removed from the creek on February 13, 2018 and data downloaded. Exact time and dates of deployments and downloads for each logger are provided in the spreadsheet data under the metadata tabs (WaterTempLoggerData_Exported_FlatCreek_2017-2018.xlsx) and the overview spreadsheet (WaterTemperatureLogger_Deployment&DownloadInfo).

Instantaneous spot water temperature measurements were taken at select locations (e.g. Cache Creek outlet, below thaw wells, thaw well outlets, etc.) during monitoring events. A Cole-Parmer DigiSense (Model 90000-29) thermometer was used to measure water temperature. The thermometer has a Traceable Certificate of Calibration for Hi-Accuracy Narrow Range Thermometers (4244-7168352).



Photo 1A & 1B. Temperature logger encased in steel tube, attached to anchor via chain. Level logger similar.



Photo 2. Temperature and level logger deployment in creek.

Air Temperature

Air temperature was downloaded from MesoWest for the Forest Service weather station at Jackson (ID: JKNW4), with Latitude 43.48556, Longitude: -110.76139, Elevation: 6240 ft, for the sampling period.

Stream Stage

Stream stage or water level was measured at eight (8) sites as shown and described in Appendix A. Staff gages were attached to a fence t-post placed in the water at the edge of the channel. Photo 3 provides a visual of the typical staff gage location relative to the stream channel. The staff gage measurements were recorded in the *Collector for ArcGIS* geodatabase during monitoring events. Notes were taken on whether the measurement was of water, ice, or snow.



Photo 3. Staff gage installed in Flat Creek. Temperature logger fence t-post on left.

Water Level Loggers/Pressure Transducers (Continuous)

HOB0 U20L-04 level loggers were installed at six (6) sites in Flat Creek and 1 in ambient air on November 7-8, 2017 as described and shown in Appendix A. Continuous measurements of both temperature in Celsius and absolute pressure in kPa were measured and logged every 5 minutes. The loggers were protected with a galvanized steel conduit attached by a chain to an anchor placed mid-channel on the bed of the creek. The anchor was connected to a fence t-post on the bank. Where located with a water temperature logger, the two loggers were placed on the same anchor but attached with separate chains.

Water level loggers were downloaded and redeployed mid-season on December 13, 2017 and removed from the creek on February 13, 2018 with the final data downloaded. Dates and times of deployments and downloads for each logger are provided in the overview spreadsheet (WaterLevelLogger_Deployment&Download.xlsx). When redeployed, the same location of loggers was attempted but may have varied slightly.

Stream Discharge

Supplemental stream discharge measurements were taken below Thaw Well #2 to provide data when the USGS Flat Creek gaging station was shut down due to ice conditions. Discharge was measured using the wading method described in Rantz et al (1982), *Measurement and Computation of Streamflow: U.S. Geological Survey Water-Supply Paper 2175, 2 v.* A Hach FH950 flow meter was used to measure water velocity. Border and shelf ice conditions reduced the accuracy of the measurements. Photo 4 shows a velocity measurement being taken through a hole drilled in the surface ice during winter 2015-2016.



Photo 4. Water velocity measurement taken through shelf ice.

Time-lapse Photographs

Time-lapse digital cameras were set up at seven (7) sites along Flat Creek, to detect changes throughout the day that would not be detected during the twice-weekly monitoring events. The cameras were set up to take photographs every 15 minutes during the daylight hours. Memory cards were downloaded from the cameras approximately every two weeks. When necessary, snow was removed from in front of the cameras, in order to maintain the best possible view of the creek.

Spot Photos

Georeferenced photographs were taken during monitoring events at “spot” locations of interest along the monitoring event routes. Spot photographs were taken using the tablet and recorded in the *Collector for ArcGIS* geodatabase during monitoring events. Photos are linked to points in the geodatabase as well as provided as jpeg image files by date on the DVD.

Ice Formation/ Flooding

New for the 2017-2018 winter monitoring, ice dam formations were more intensely documented. The ice dams were uniquely identified, additional attributes collected (Table 1), and photos taken. They were then re-visited during subsequent monitoring events to document ice dam progression and regression.

Ice formation and flooding characteristics were mapped on the Tablet using lines, polygon and point shapefiles in the *Collector for ArcGIS* geodatabase. The Data Summary section below describes each of the data types and attributes collected (Table 1). After each monitoring event the geodatabase was downloaded from the Tablet, errors in the data were fixed and the daily geodatabases combined.

Weather

Weather conditions during monitoring events were recorded at the start and end of monitoring events. Since the monitoring events lasted approximately 3-4 hours and typically revolved around sunrise, air temperature, shade and cloud conditions changed significantly from start to finish. The weather data were recorded in the field on the tablet as a point within the *Collector for ArcGIS* geodatabase.

DATA SUMMARY

Data from the collection period (11/12/17 to 02/13/18), individual monitoring events, and data downloaded from external sources are provided in digital format on the DVD provided with this document and in an *ArcGIS Online Viewer*. The following describes the digital data and how to access it.

ArcGIS Geodatabase & Online Viewer

An Esri ArcGIS geodatabase was created for the project and contains data collected using *Collector for ArcGIS* (Photo 5) in the field by Alder Environmental as well as water temperature loggers, water level loggers, time-lapse cameras, and thaw well locations. Table 1 below provides a summary of the geodatabase features and attributes. These data are intended to be used to sort, display and research data by date and time as well as to create maps for different scenarios.

The digital deliverables for the ArcGIS Geodatabase are provided through two methods:

1. ArcGIS geodatabase (FCWID_2017_2018_FINAL_180306.gdb) and ArcMap project file (FCWID_2017_2018_FINAL_180306.mxd) with full functionality of the user's license level is contained on the DVD and
2. *ArcGIS Online Viewer* which allows for viewing of data, attributes, and photo attachments through a website but not editing of the geodatabase information.

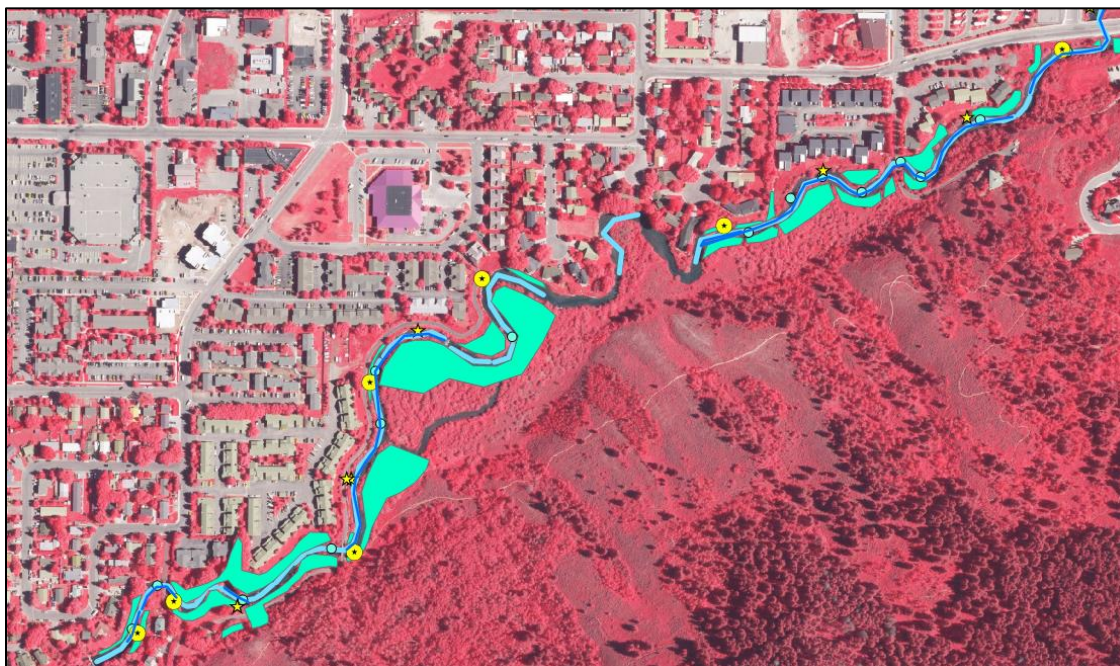


Photo 5. Screenshot of monitoring event geospatial data collected and recorded in *Collector for ArcGIS*.

Table 1. ArcGIS Geodatabase layer and attribute matrix for Flat Creek Winter 2017-2018 monitoring.

Layer \ Attribute	Backwater in Channel	Flooding Overbank	Ice Anchor	Ice Dam	Ice Frazil	Ice Overbank	Ice Surface	Open Water Lead	Spot Photo	Staff Gage	Water Spot Temperature	Monitoring Event Weather	Water Temp Logger Location	Water Level Logger Location	Timelapse Photo Camera Locations	Thaw Well Location	Areas Not Observed
Feature Type	—	□	—	•	•	□	—	—	•	•	•	•	•	•	•	•	—
Date Time	X	X	X	X	X	X	X	X	X	X	X	X					
Estimated Height	X			X					X								
Structures		X															
Type			X														
Percent Width				X													
Source				X													
Activity *				X													
Catalyst *				X													
Notes				X						X	X						
Density					X												
Thickness						X											
Estimated Percentage							X										
Estimated Width								X									
Mountain Shadow									X	X							
Description									X								
Direction									X								
Feature Type									X								
Height										X							
Temperature											X						
Location											X						
Air Temperature												X					
Cloud Cover												X					
Precipitation Current												X					
Precipitation 12 Hours												X					
Wind												X					
Staff Gage ID										X							
Site Name													X	X	X	X	
Attachment (Photo) *				X					X								
FEATURE TYPE KEY • point — line □ polygon																	

* New attributes for Ice Dams 2017-2018.

ArcGIS Geodatabase

Data is provided as two items in the GIS Geodatabase folder on the DVD:

- FCWID_2017_2018_FINAL_180306.mxd
- FCWID_2017_2018_FINAL_180306.gdb

To work with these files on a computer (rather than the DVD), copy the entire folder onto your computer. The .mxd file contains relative paths. Metadata can be viewed within ArcCatalog.

ArcGIS Online Viewer

For access to ESRI *ArcGIS Online Viewer*, the following information is required:

Weblink: <http://arcg.is/1m1aHf> (Previous winter data is included in this link as well)

Username: FCWID_2016

Password: frazil_2018

(both are case sensitive)

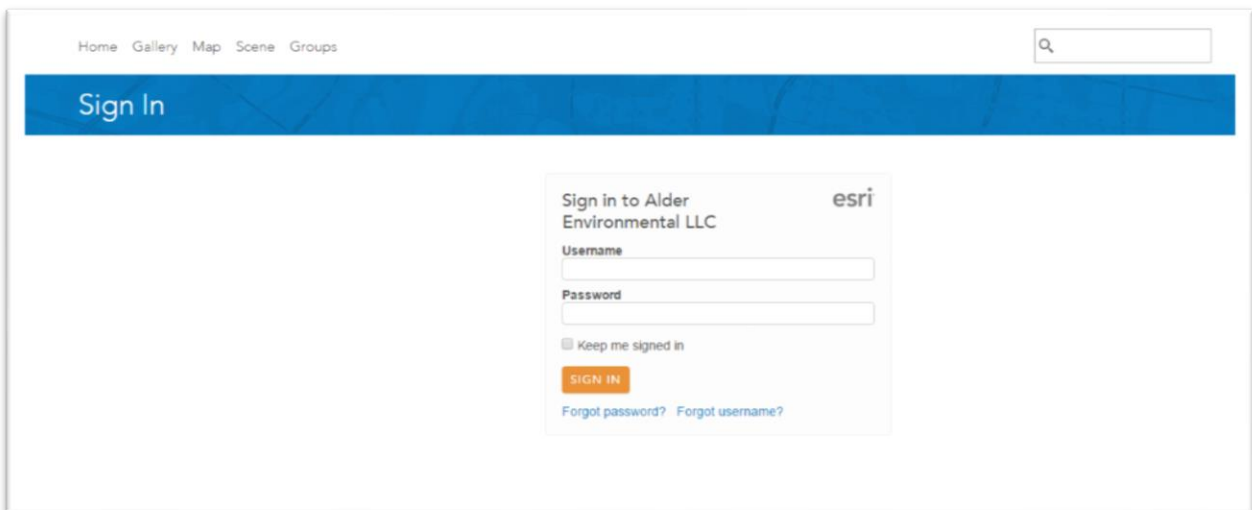
Email account: flatcreek@fcwid.org

Security question: What is your high school mascot? Answer: Broncos.

The email and security question are only needed in the case of resetting the password.

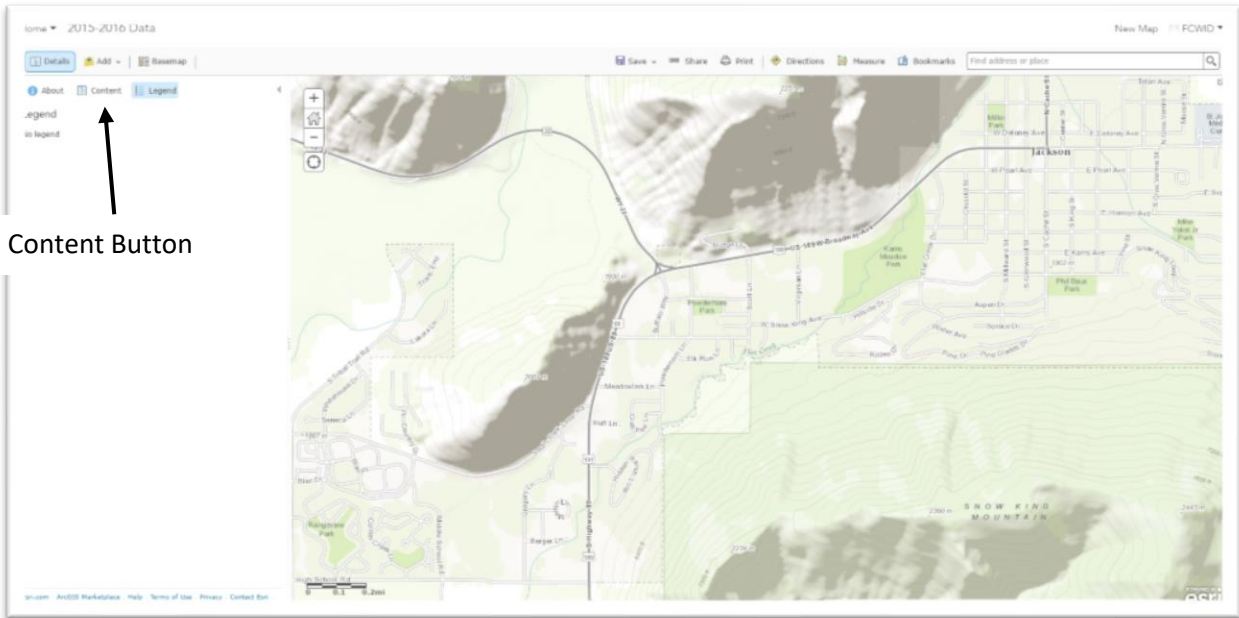
Basic Online Viewer Instructions:

1. Type the weblink above into a browser. A sign-in page with Alder Environmental, LLC as a heading will appear.

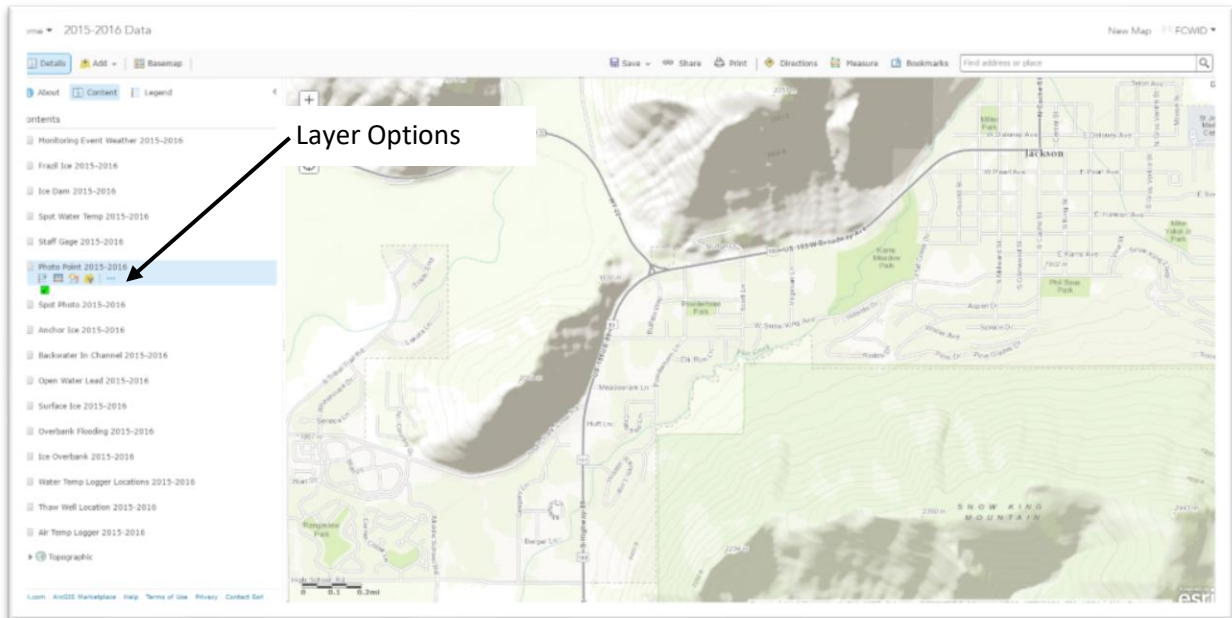
The screenshot shows a web browser window displaying the sign-in page for the ArcGIS Online Viewer. At the top, there is a navigation bar with links for Home, Gallery, Map, Scene, and Groups, along with a search icon. Below this is a blue banner with the text "Sign In". The main content area features a sign-in form titled "Sign in to Alder Environmental LLC" with the Esri logo. The form includes fields for "Username" and "Password", a "Keep me signed in" checkbox, and a "SIGN IN" button. Below the button are links for "Forgot password?" and "Forgot username?".

2. Sign in with the username and password listed above.

3. A map page titled “Flat Creek Winter Characterization Study Data” will open. By default, the winter 2017-2018 data is turned on. Previous winter study data is accessible via the Table of Contents.



4. Click on the Content button at the top left and a Table of Contents will appear.



5. The Table of Contents allows you to turn layers on and off (check box to the left of the layer name), view the attribute table for each layer and filter (aka query) layers by attributes (dates, fixed points, etc.).

Legend  (adds symbology for layer to the table of contents view)

Filter  (opens a window to query data to be viewed for that layer)

Table  (displays the layer’s attributes in table format)

ArcGIS Online has excellent online help resources to assist with learning to use the map interface (<https://doc.arcgis.com/en/arcgis-online/use-maps/view-maps.htm>). Your *ArcGIS Online* status with this map is as a User. Therefore, capabilities may be limited to viewing and querying the data set. Users should not be able to edit the data set.

Ice Formation: Ice Dams

The ice dams were uniquely identified when they newly formed and subsequently monitored throughout the rest of the monitoring season. Additional to previous attributes, new attributes catalyst, activity, and photos were also collected (Table 1) and are all accessible through the ArcGIS Geodatabase. The purpose was to intensely monitor the ice dams during this monitoring season. In particular, the photos provide a visual timeline of the progression and regression of the individual ice dams.

Ice dam photos taken during monitoring events are provided in two locations:

1. Within the Ice Dam Summary (FCWID_IceDamsSummary_2017-2018.xlsx) located in the Monitoring Photos&Ice Dams\Ice Dams folder. This Excel file has photo links to the .jpg files in dated subfolders of the Monitoring Photos&Ice Dams\Ice Dams folder in the DVD and
2. As georeferenced photos associated with uniquely identified points and ice dam attributes in the geodatabase.
 - a. To streamline the geodata access to all the individual ice dam monitoring events, data for each unique ice dam was stacked onto one single point with the ability navigate through all the various monitoring events for each ice dam.

Photos 6-8 below provide an example of the visual timeline of the progression and regression of the individual ice dams.



Photo 6. Ice Dam #1 newly formed on December 12, 2017



Photo 7. Ice Dam #1 in progression (active) on December 28, 2017



Photo 8. Ice Dam #1 in regression (absent) on January 16, 2018

Spot Photos

Photos taken during monitoring events are provided in two locations:

1. As .jpg files in dated subfolders of the Monitoring Photos&Ice Dams\Spot Photos folder in the DVD and
2. As georeferenced photos associated with points in the geodatabase.

Time-Lapse Photos

Time-lapse photos taken by the seven (7) trail cameras are provided as .jpg files in subfolders labeled by camera (C17.1 through C17.7) within the Monitoring Photos_Ice Dams\Timelapse Photos folder in the DVD. All photos are labeled numerically with the naming convention given by the camera. Video software may be used to view the photos sequentially.

Note: Unfortunately, there was a technical issue during the download of photos from some of the cameras and photo data was lost. Cameras C17.1 through C17.3, and C17.7 are missing a portion of the monitoring season.

Missing Photo Data:

C17.1 – C17.3: November 14 – December 12, 2017

C17.7: December 14, 2017 – January 28, 2018

Landowner Photos

One landowner, Franz Camenzind, provided photos of their observations. These photos are provided on the DVD under the folder Monitoring Photos_Ice Dams\Landowner Photos.

Water Temperature (loggers)

Water temperature logger data (369,296 data points from Flat Creek) are provided in two digital formats in the Water Data\Water Temp Logger Data Folder on the DVD:

1. Excel spreadsheets (.xlsx)
 - a. WaterTempLoggerData_Exported_FlatCreek_2017-2018.xlsx [2 tabs for each logger (metadata & data)]
 - b. WaterTempLoggerData_Exported_IceBath_2017-2018.xlsx [2 tabs for each logger (metadata & data)]
 - c. WaterTemperatureLogger_Deployment&DownloadInfo.xlsx
2. RBR Ruskin raw data download files (.rsk)
 - a. File Naming Convention:
LoggerID_LoggerSerialNumber_DateOfDownload_TimeOfDownload
 - b. For each logger there are 3 files by date:
 - i. 20171106 - initial ice bath calibration
 - ii. 20180213/14 - data download
 - iii. 20180215 - final ice bath

Air Temperature & Precipitation

Air temperature and precipitation data was downloaded from the MesoWest weather station (JKNW4) online database and is provided as an Excel spreadsheet within the folder Air Temps on the DVD provided with this document.

- AirTemp_JKNW4_WeatherStation_171112-180213.xlsx

Water Level (loggers)

Note:

During 2nd deployment, loggers were replaced as close as possible to original location/depth. However, there were un-intended variations in the replacement location/depth. The data will indicate this variation during 2nd redeployment period.

Water level logger data (171,585 data points from Flat Creek) are provided in three digital formats in the Water Data\Water Level Logger Data Folder on the DVD:

1. Excel spreadsheets (.xlsx)
 - a. WaterLevelLoggerData_BarometricCompensated_Exported_FlatCreek_2017-2018.xlsx
 - b. WaterLevelLogger_Deployment&DownloadInfo.xlsx
2. HOBOWare raw data download files (.hobo)
 - a. File Naming Convention:
LoggerID_DateOfDownload
 - b. For each logger [L17.1 through L17.7], there are 2 files by date:
 - i. 171213 - first data download
 - ii. 180213 - second data download
3. HOBOWare project files for processed (barometric compensated) data (.hproj)
 - a. File Naming Convention:
LoggerID_”Merged_BarometricCompensated”

Stream Discharge (instantaneous)

A total of four (4) instantaneous stream discharge measurements taken within Flat Creek are summarized and provided in detail within an Excel spreadsheet located in the Water Data folder on the DVD (Stream_Discharge_2017-2018.xlsx).

Thaw Wells

Data for thaw wells included operations, water temperature and estimated flow rates. This information is compiled in an Excel spreadsheet in the Water Data folder on the DVD (Thaw Well Data & Operations_2017-2018.xlsx). The spot water temperature measurements of the thaw well discharges are also included in the geodatabase.

USGS Gage Data

Data downloaded from the USGS 13018350 FLAT CREEK BELOW CACHE CREEK, NEAR JACKSON, WY stream gaging site is provided in the Water Data folder on the DVD in an Excel spreadsheet (USGS Flat Creek Gage Data Surface & Well_2017-2018.xlsx). Data downloaded include:

Flat Creek

Discharge, cubic feet per second

Gage height, feet

Stream water level elevation above NAVD 1988, in feet

Temperature, water, degrees Celsius

Groundwater

Elevation above NAVD 1988, feet

Temperature, water, degrees Celsius

Depth to water level, feet below land surface

APPENDIX A – MONITORING PARAMETERS AND SITES

Sheet 1 - Monitoring Parameters & Data

Sheet 2 - Monitoring Sites

Figure 1 - Monitoring Sites

Figure 2 - Monitoring Sites North

Figure 3 - Monitoring Sites Middle

Figure 4 - Monitoring Sites South

FLAT CREEK WINTER CHARACTERIZATION STUDY 2017 - 2018

SHEET 1 of 2 - Monitoring Parameters & Deliverables

Prepared for: Flat Creek Water Improvement District
9/5/2017



PEER REVIEW: Ed Kempema, Ph.D. (Geology and Geophysics Dept., Uni. Of WY) provided recommendations regarding this monitoring plan. (Aug. 24, 2017 phone correspondence / Sept. 5, 2017 email correspondance)

LOCATION REFERENCE: Refer to Sheet 2 & Figures 1-4

DATA COLLECTION PERIOD: November 12, 2017 – February 13, 2018

Data Parameter	Sites (IDs)	Source/Method	Frequency	Description	Deliverable
Water Temperature	13 (T17.1 thru T17.13)	RBR soloT loggers	5 minutes, downloaded every ~30 days	Accuracy: ±0.002 C, Resolution: <0.00005 C	Spreadsheet
	USGS Gaging Station	Website	downloaded end of February 2018		Spreadsheet
	Multiple (R-1a, R-1, R-2, R-3)	Field thermometer	Approx. 2 times per weeks		Point GIS shapefile
Air Temperature	Jackson, WY (JKNY)	Website	downloaded end of February 2018	Bureau of Rec. station at BTNF Headquarters	Spreadsheet
Precipitation	Jackson, WY	Website	downloaded end of February 2018	Bureau of Rec. station at BTNF Headquarters	Spreadsheet
Stream Discharge	1 (Crab Tree Bridge)	Field measurement	6 times, as needed to supplement USGS	Area/velocity method	Spreadsheet
Stream Stage	8 (S17.1 thru S17.8)	Field, staff gage	Approx. 2 times per weeks	To document water level rises due to ice	Spreadsheet
	USGS Gaging Station	Website	downloaded end of February 2018		Point GIS shapefile
Water Level	7 (L17.1 thru L17.7)	HOBO U20L-04 level logger	5 minutes, downloaded every ~30 days	Accuracy: ±0.4 cm water, Resolution: <0.14 cm	Spreadsheet
Groundwater Temp/Depth	Thaw Wells #1,2,3	Field measurement	Once, possibly from as built logs	Depth to groundwater and water temperature	Spreadsheet & Charts
	USGS Gaging Station	Website	downloaded end of February 2018	Water temperature	Spreadsheet
Thaw Wells	Thaw Wells #1,2,3	Field msrmt./Recordings	As needed	Measure water temp. / flows, record operations	Spreadsheet
Shade/Sunlight/Cloud Cover	District	Observations/Recordings	Approx. 2 times per weeks	Recorded on map and field notes	Monitoring event maps & Field Notes; GIS Shapefiles
Snow King Snowmaking	District	Observations/Recordings	Approx. 2 times per weeks	Drift to creek recorded on map and field notes	Monitoring event maps & Field Notes
Ice Formation/ Flooding	District, Routes (R-1a, R-1, R-2, R-3)	Observations/Recordings	Approx. 2 times per weeks <u>Observation Triggers</u> (general guidelines) • 2 consecutive days of air temp <32°F or • 1 consecutive days of air temp <25°F or • air temp <10°F or • Observed ice formation in creek	Record and map observations at least twice a week and during prolonged high pressure cold periods; goal is to record conditions on 75% of creek length such as ice type, location and thickness e.g. frazil, anchor, shelf, jam, entrained/floating, etc.; width of open water downstream of thaw wells; USACE Ice Observation Report Form content	Monitoring event maps, GIS shapefiles & Field Notes; Maps will include date, time, morning air temperature & cloud cover, location of ice dams, area of backwaters upstream of any ice dams, open water leads downstream of thaw wells (if they are in operation), any other open-water leads, and locations of any impacted (flooded) areas and structures. Completed USACE Ice Observation Reports.
Timelapse Photos	7 (C17.1 thru C17.7)	Digital Camera	15 minutes, downloaded every 2 weeks	Upstream	Photo log, digital files, Point GIS shapefile
Photos, Spot	Ice Dams	Digital Camera w/ GPS	As needed	Area view and close up of ice type	Photo log, digital files, Point GIS shapefile
Summary Report	All data collected will be delivered in digital formats.				Digital copies; GIS geodatabase & shapefiles;
FEATURE TYPE KEY	Deliverables will be similar to the Flat Creek Winter Characterization Study Methods & Data Summary Data Deliverables Report (April 8, 2016)				ESRI ArcGIS Online

FLAT CREEK WINTER CHARACTERIZATION STUDY
SHEET 2 of 2 - Monitoring Sites

2017 - 2018



Prepared for: Flat Creek Water Improvement District
9/5/2017

Refer to Figures 1-4 for Location Reference

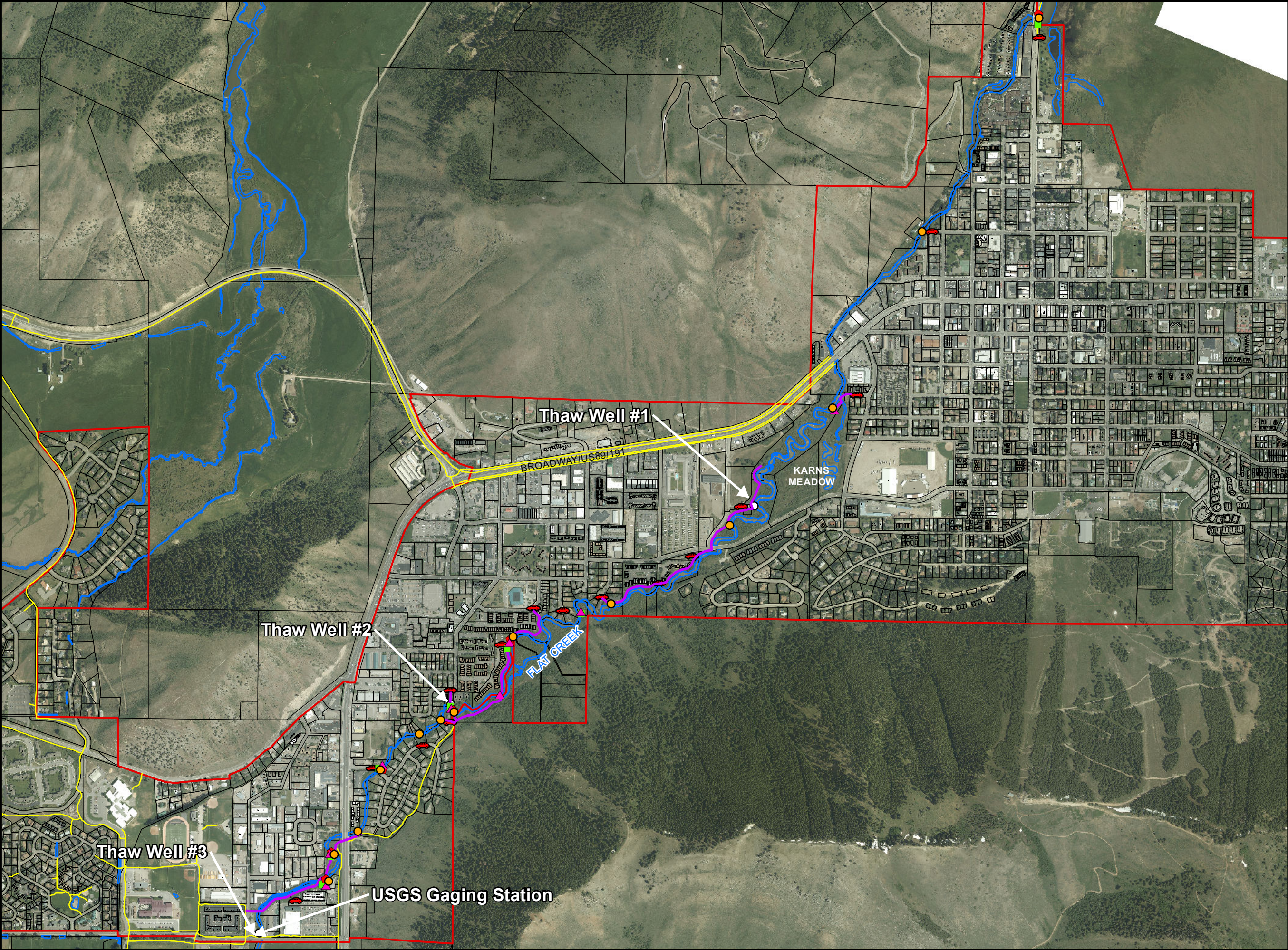
WATER TEMPERATURE (loggers)			
Site ID	Description	Property Ownership	General Location
T17.1	Upstream of Town (shelf ice)	USFWS/WYDOT ROW	Upstream of N. Cache/Highway bridge
T17.2	Ice Reach (steeper gradient starts)	Town/Lukas Farms LLC	West end of Gill St. downstream of bridge
T17.3	Below Kelly Tube/Cache Creek	Town of Jackson	Karns Meadow (Hansen St.)
T17.4	Below Thaw Well #1 (mixed)	Town of Jackson	Karns Meadow (Town Pump House)
T17.5	Ice Reach	Camenzind	480 Stacey Lane (Camenzind)
T17.6	Ice Reach (Above rock weir)	Town/Creekside Village	Garaman Pathway/East of Creekside Vill.
T17.7	Above Thaw Well #2	Town/Flat Creek Condos	Garaman Park
T17.8	Below Thaw Well #2 (mixed)	Buckstaff/Stanley	300 Hidden Spur Lane
T17.9	Ice Free Reach	Town of Jackson	2110 Hidden Ranch Lane (Vacant Lot)
T17.10	Ice Free Reach	Town of Jackson	Stellaria Lane bridge
T17.11	Ice Reach	Greg Prugh	Just upstream of Broadway/US89/191 bridge
T17.12	Ice Reach	Town of Jackson	Just upstream of bike path bridge
T17.13	Ice Reach	Eagle Village Condos	Just upstream Eagle Village bike path bridge
USGS	Ice Reach	Town of Jackson	USGS Gaging Station @ High School Rd.

STREAM STAGE (staff gages)			
Site ID	Description	Property Ownership	General Location
S17.1	Below Kelly Tube/Cache Creek	Town of Jackson	Karns Meadow (Hansen St.)
S17.2	Ice Reach (across creek)	Klynn	Dogwood Lane (Klynn)
S17.3	Ice Reach (Above rock weir)	Town/Creekside Village	Garaman Pathway/East of Creekside Vill.
S17.4	Ice Reach	Creekside Village	East of Creekside Village @ Garaman Path
S17.5	Above Thaw Well #2	Town/Flat Creek Condos	Garaman Park
S17.6	Ice Free Reach	Town of Jackson	Stellaria Lane bridge
S17.7	Ice Reach	Town of Jackson	Just upstream of bike path bridge
S17.8	Ice Reach	Eagle Village Condos	Just upstream Eagle Village bike path bridge
USGS	Ice Reach	Town of Jackson	USGS Gaging Station @ High School Rd.

WATER LEVEL (loggers)			
Site ID	Description	Property Ownership	General Location
L17.1	Upstream of Town (shelf ice)	USFWS/WYDOT ROW	Upstream of N. Cache/Highway bridge
L17.2	Ice Reach (Above rock weir)	Town/Creekside Village	Garaman Pathway/East of Creekside Vill.
L17.3	Above Thaw Well #2	Town/Flat Creek Condos	Garaman Park
L17.4	Ice Free Reach	Town of Jackson	Stellaria Lane bridge
L17.5	Ice Reach	Town of Jackson	Just upstream of bike path bridge
L17.6	Ice Reach	Eagle Village Condos	Just upstream Eagle Village bike path bridge
L17.7	Barometric control logger	Private residence	In garage, in Town of Jackson

TIMELAPSE PHOTOS (upstream)			
Site ID	Description	Property Ownership	General Location
C17.1	Upstream of Town (shelf ice)	USFWS/WYDOT ROW	Upstream of N. Cache/Highway bridge
C17.2	Ice Reach (across creek)	Klynn	Dogwood Lane (Klynn)
C17.3	Ice Reach	Elk Run	South of Elk Run @ Garaman Path
C17.4	At Thaw Well #2	Town/Flat Creek Condos	Crabtree Path Bridge
C17.5	Ice Free Reach	Town of Jackson	Stellaria Lane bridge
C17.6	Ice Reach	Town of Jackson	Just upstream of bike path bridge
C17.7	Ice Reach	Eagle Village Condos	Just upstream Eagle Village bike path bridge

WALKING ROUTES (spot temperature, photos, and GPS ice dams)			
Site ID	Description	Property Ownership	General Location
R-1a	Below Kelly Tube/Cache Creek	Town of Jackson	Karns Meadow (Hansen St.)
R-1	Above to Below Thaw Well #1	Mixed	Karns Meadow to 810 West
R-2	Above Thaw Well #2	Pathway	Elk Run Ln. to Crabtree Ln.
R-3	Ice Area near Smiths/Berger Ln.	Pathway	South End of Town along Pathway



Flat Creek Water Improvement District

FLAT CREEK
WINTER
CHARACTERIZATION

Winter 2017 - 2018
Jackson, WY

FIGURE 1:

OVERVIEW
Monitoring Sites

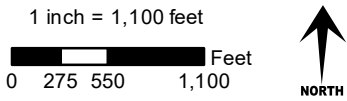
September 5, 2017

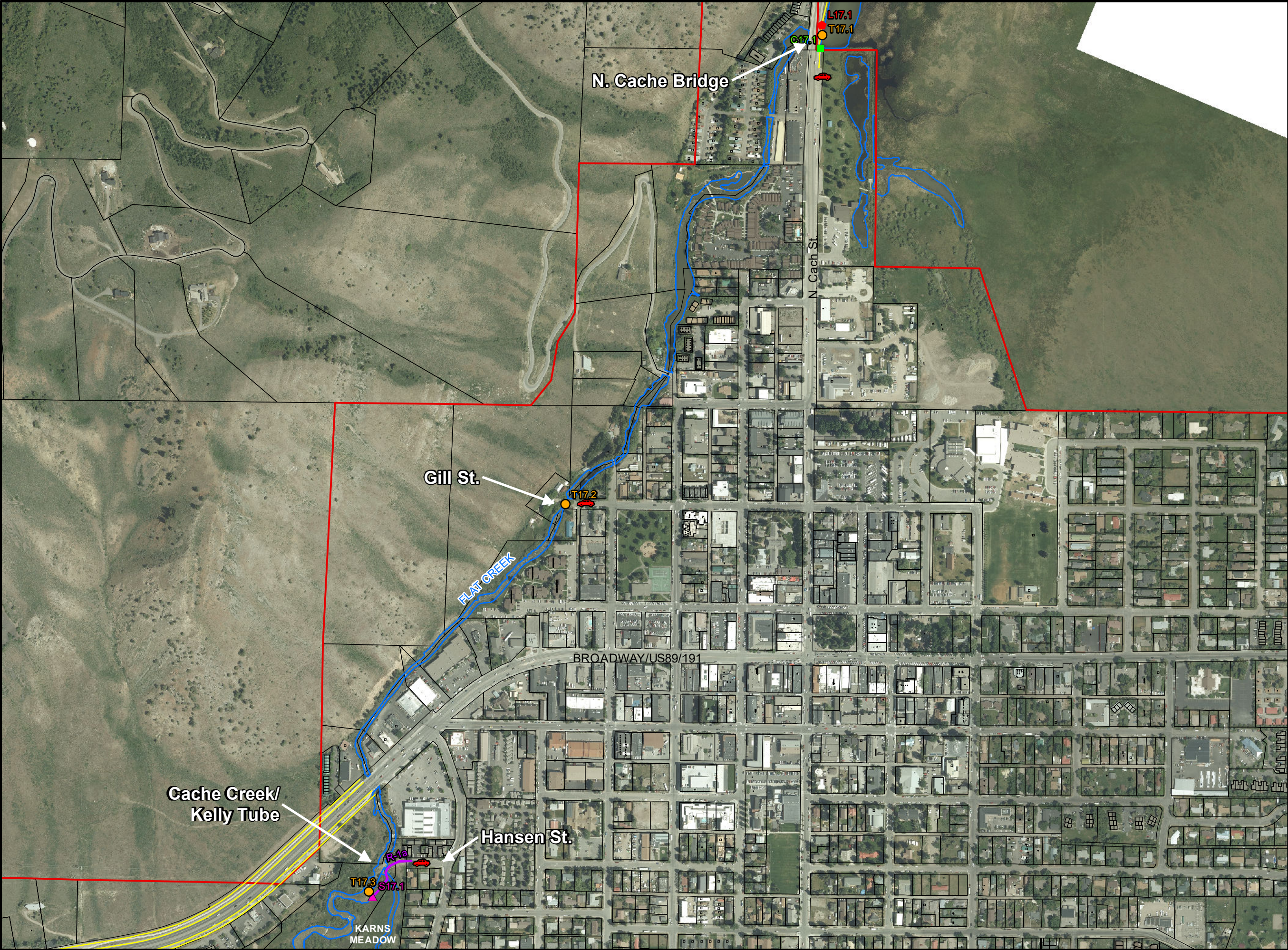
Legend:

Monitoring Sites

- Water Temp. Logger
- Staff Gages
- Water Level Logger
- Timelapse Photos
- Walking Routes
- Parking Locations
- Thaw Well
- Pathway
- Town of Jackson
- Flat Creek
- Property Boundary

2017 Aerial Imagery





Flat Creek Water Improvement District

FLAT CREEK
WINTER
CHARACTERIZATION

Winter 2017 - 2018
Jackson, WY

FIGURE 2:

NORTH
Monitoring Sites

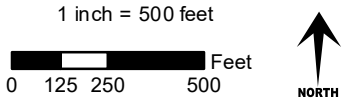
September 5, 2017

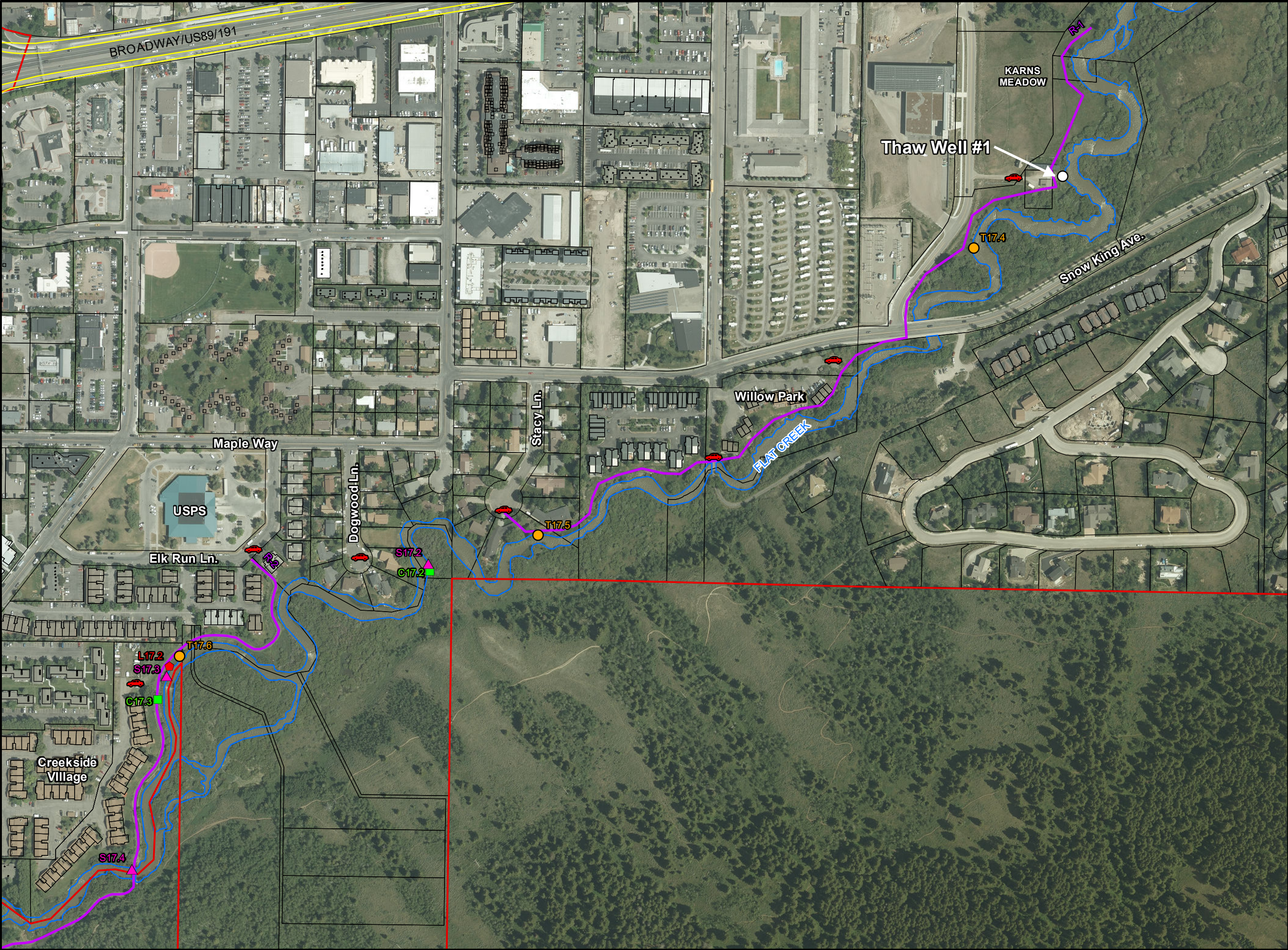
Legend:

Monitoring Sites

- Water Temp. Logger
- Staff Gages
- Water Level Logger
- Timelapse Photos
- Parking Locations
- Walking Routes
- Thaw Well
- Pathway
- Town of Jackson
- Flat Creek
- Property Boundary

2017 Aerial Imagery





Flat Creek Water Improvement District

FLAT CREEK
WINTER
CHARACTERIZATION

Winter 2017 - 2018

Jackson, WY

FIGURE 3:

MIDDLE
Monitoring Sites

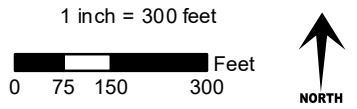
Septemeber 1, 2017

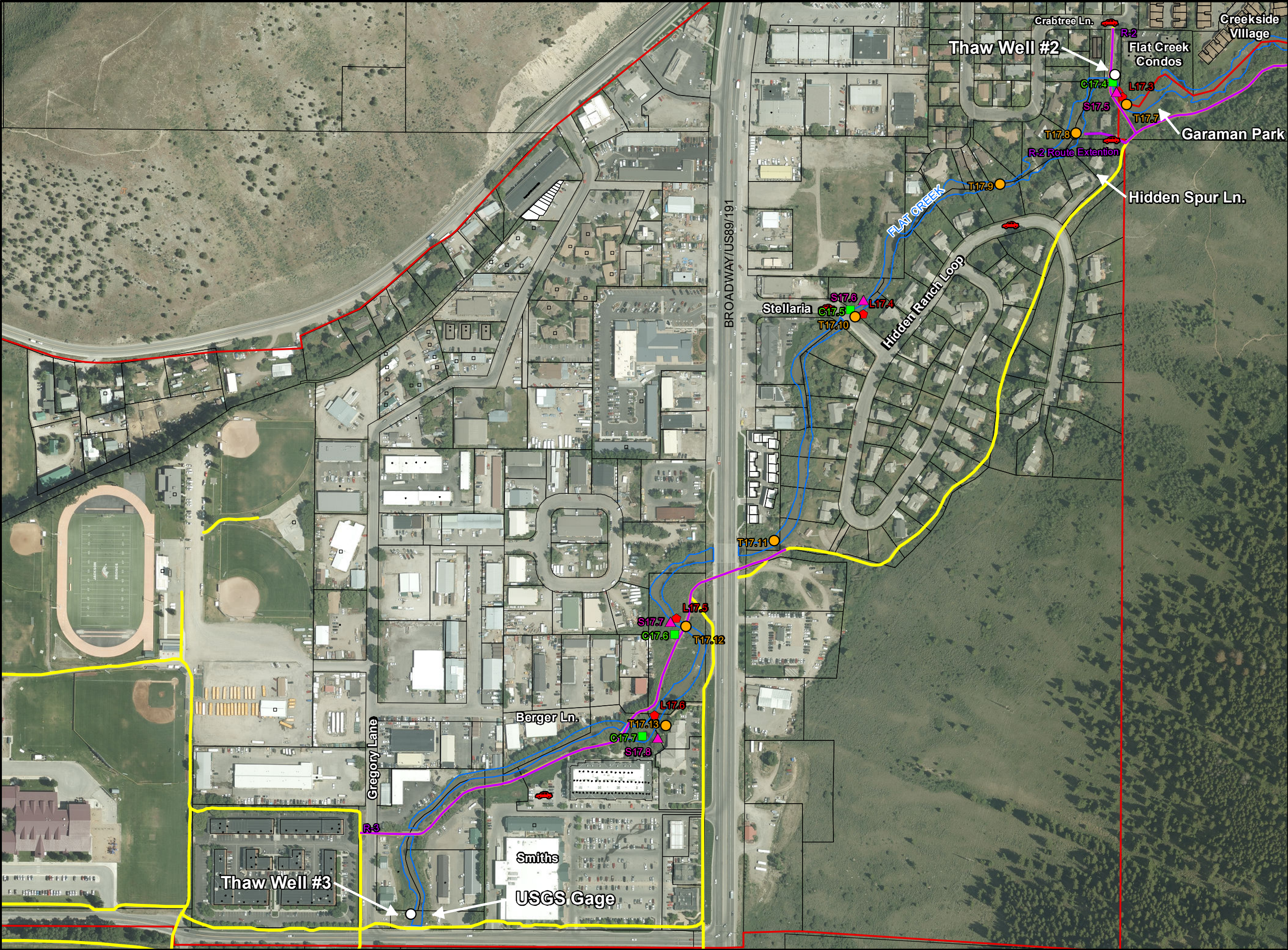
Legend:

Monitoring Sites

- Water Temp. Logger
- Staff Gages
- Water Level Logger
- Timelapse Photos
- Walking Routes
- Parking Locations
- Thaw Well
- Pathway
- Town of Jackson
- Flat Creek
- Property Boundary

2017 Aerial Imagery





Flat Creek Water
Improvement District

FLAT CREEK
WINTER
CHARACTERIZATION

Winter 2017 - 2018

Jackson, WY

FIGURE 4:

SOUTH
Monitoring Sites

September 5, 2017

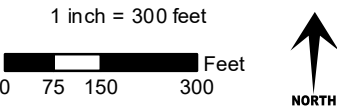
Legend:

Monitoring Sites

- Water Temp. Logger
- Staff Gages
- Water Level Logger
- Timelapse Photos
- Walking Routes
- Parking Locations
- Thaw Well

- Pathway
- Town of Jackson
- Flat Creek
- Property Boundary

2017 Aerial Imagery



ALDER ENVIRONMENTAL, LLC
Water • Wetlands • Ecological Consulting
P.O. Box 6519, Jackson, Wyoming 83002
(307) 733-5031 www.alderenvironmental.com